D2R PCVR: What We Added

Contributions beyond the original BodyWalkVR D2R VR codebase

Snapshot: October 11, 2026 • Windows preview v0.3.3 • Source 2e65671b396966e558f2f716f65f577e3b735d16

This document inventories the additions and changes in our adaptation. It distinguishes inherited capabilities from our work, and separates implemented features from headset confirmation and experimental options. The current preview targets Steam Diablo II: Resurrected 3.3.93787 and local offline characters.

Foundation and attribution

Our codebase starts from natpoh/diablo2-vr-port (BodyWalkVR's D2R VR), at commit 10cfb0a2a9080184101794d3ac8c238c35717967. Its MIT license and copyright notices are retained. The original emmericp/D2R-3D project was a reference; its checkout is separate, and it is not our direct source-code base.

The upstream mod already supplied the first-person camera, character body and controller-aim foundation, stereo scene drawing, D2RLoader integration, ReShade sky/fog/ceiling effects, and camera/draw-distance facilities. Its VR route used BodyWalk and FlatVR. We adapted that foundation rather than writing the entire mod from scratch.

Diablo II: Resurrected must be purchased and installed separately. Neither our repository nor the release ZIP supplies Blizzard's game executable, models, textures or audio. Native icons and potion models are read from each player's own installation into local caches. Other licensed components retain their own notices.

1. Native VR runtime and headset support

- Added a native OpenXR transport using the game's D3D12 device and graphics queue. It manages the XR session, predicted poses, per-eye views/projections, swapchains, image copies and compositor submission.
- Replaced the runtime dependence on BodyWalk/FlatVR with direct OpenXR rendering and controller input. The current Windows release does not require those products or Proton/Wine.
- Added tracked headset, hand grip and aim poses; trigger/grip/stick/face-button input; controller haptics; recentering; and release/recovery behavior across focus or tracking changes.
- Added runtime/session capability reporting and guarded fallbacks for unsupported graphics formats, composition-layer limits, gaze support and variable-rate shading.
- Added native Windows build support and retained the earlier Linux/Proton development path. Linux work is historical context; this release and its player instructions focus on Windows.

Headset / route	What we added and verified
Steam Frame / SteamVR	Direct native Windows OpenXR path; offline gameplay confirmed. Steam Frame uses SteamVR streaming.

Headset / route	What we added and verified
Quest 3 / Steam Link + SteamVR	Touch controller bindings and Windows launch routing; offline gameplay confirmed.
Quest 3 / Virtual Desktop + VDXR	Explicit per-launch VDXR selection without SteamVR; gameplay and corrected colors confirmed.
Quest 2	Quest controller mappings and setup choice are included; no physical Quest 2 verification yet.
Virtual Desktop + SteamVR	Distinct from the confirmed VDXR route; this combination has not been separately verified.

2. Stereo rendering and visual corrections

- Added per-eye native projection handling, plus a centered-projection compatibility option for the game's lighting bins. This addresses view-dependent lighting patches while retaining tracked eye positions; a wider rendered field of view can soften angular pixel density.
- Reworked creature name/health label projection and current-eye draw ordering so native labels use the correct eye's anchor and tooltip update. The combined ordinary-creature fix was confirmed on Steam Frame; broader boss and scene coverage remains limited.
- Added stable shared-eye billboard handling and narrowly scoped town-portal corrections. Both tested portal endpoints were confirmed aligned after correcting their center and facing transforms; other cosmetic variants are not exhaustively covered.
- Added native depth-range and motion/depth compatibility work for particles and the experimental DLSS path. Smoke/effects clarity improved in owner testing, but DLSS still has unresolved clarity issues and remains off in the recommended preset.
- Added UI-only extraction/conversion and a shared menu image for both eyes, rather than accidentally displaying a separate world-contaminated UI capture to each eye.
- Fixed the asymmetric black/transparent background in management panels. Stash, inventory, cube, mercenary and trade-related panels now use the shared opaque UI policy. Matching stash backgrounds were confirmed in the headset.
- Fixed typed texture-view handling for the typeless resources exposed by SteamVR/VDXR. This restored potion/UI rendering on native Windows and retains the intended sRGB interpretation.
- Fixed Virtual Desktop's washed-out colors by choosing an sRGB OpenXR swapchain for the game's encoded SDR image, even when VDXR also offers linear UNORM. Normal colors and contrast were confirmed.
- Fixed the installer's malformed camera INI section, which caused Windows to ignore the intended near clip and hide the torso, arms and parts of the shield close to the head. Correct torso/arm/shield visibility was confirmed after correcting the setting.
- Added native-VR first-person mode repair/guards so old saved third-person modes cannot silently break the VR camera. Flat-screen camera modes remain separate.

3. Performance work and frame pacing

These are implemented optimizations or configurable experiments, not a claim of a measured overall Windows speedup. The owner reports very good Windows performance, but a controlled Windows baseline-versus-current comparison has not been completed.

- Added fixed foveated rendering through D3D12 Tier 2 variable-rate shading: full-quality central vision and coarser peripheral shading for eligible opaque world passes. UI, labels, transparency, post effects and incompatible passes remain full quality.
- Added optional eye-tracked foveation using OpenXR gaze input. Immutable mask banks avoid per-frame mask allocation/upload, and invalid or stale gaze falls back to full quality. It requires supported hardware/runtime and user-enabled gaze sharing; it is off by default.
- Reduced repeated pipeline lookup work with generation-aware per-thread caches, including VRS and spatial-UI classification. Added lightweight native callback locking and bounded reusable state.
- Added guarded same-process memory-copy helpers and avoided unnecessary page-permission changes. Earlier Linux/Proton microbenchmarks demonstrated a narrower copy-path saving; that is not an equivalent game-frame saving.
- Moved change-driven wheel/wrist atlas painting to a bounded background worker. The frame thread uploads completed pixels, with stale-content guards so an old image cannot dispatch a new skill action.
- Cached portal classification within particle batches, reduced routine timing sampling, and skipped mouse-cursor queries when mouse look is inactive.
- Stopped unrelated INI reloads from triggering repeated model-visibility rebuilding. Draw-distance refreshes now follow actual relevant setting changes, startup and retry needs.
- Added separate per-eye copy command lists with paired completion handling. Also added an experimental deferred-GPU-wait option; the conservative default remains enabled through defer_gpu_wait=0.
- Added recommended installation settings: 80% resolution scale, medium shadows/ambient occlusion/atmospherics, FXAA, DLSS off, VSync off, dynamic resolution off and camera shake off. Native height fog is retained to avoid the mod’s repeated post-transition scan path. These settings involve visual tradeoffs.
- Reduced full process-tree polling in the earlier Linux launcher while retaining frequent checks of the known game process.

Performance evidence is deliberately limited: an earlier Linux gaze/full/gaze trial showed a small preliminary improvement of about 2.8% in mean stereo-cycle time. Later CPU-only comparisons did not establish an overall FPS gain. Experimental reduced draw distance and deferred waiting were rolled back after stalls/stream-quality failures. Current Windows defaults keep the conservative wait behavior; no cross-OS gain is inferred.

4. Character fit, body tracking and calibration

- Added a first-load T-pose calibration: stand upright with arms out, release both triggers, then hold both to capture a stable pose and save it.
- Added repeat calibration through the in-headset VR settings menu, with cancel/skip, save feedback, tracking/posture/floor checks and persistent settings.

- Standing eye height determines world scale; grip-to-grip span adjusts the character's upper/lower arm proportions. Controller motion remains physical 1:1, with bounded rig scaling and authored hand/equipment size retained.
- Added atomic calibration persistence and conflict handling so interrupted or conflicting saves do not silently replace a valid calibration.
- Corrected the misleading floor-setup message when the real problem was resting or bent arms. Floorless tracking spaces are still distinguished from a valid stage floor.
- Improved the body's relationship to the tracked neck, stable horizontal facing and recognized-rig leg/arm posing. Added local living-body selection so a corpse or another matching player model cannot become the tracked avatar.
- Fixed transient per-eye body gating that caused left-eye shoulder flashing. Added height-only reset and full recenter controls, with fresh session positioning so startup motion is not carried into the avatar offset.

Calibration measures standing eye height and controller span, not head-top height or fingertip wingspan. This is not a full-body tracking system; independent torso/foot trackers and exhaustive crouch/death/rig coverage are not claimed.

5. Physical melee and cave-transition recovery

- Replaced external gesture-to-button melee with a swept contact system tied to the final posed main-hand weapon. Blade motion follows translation and rotation between frames and is tested against approximate creature body capsules.
- A valid contact requests native basic-attack damage for the contacted creature on the authoritative game thread. It does not require pressing A, selecting a target first, or waiting for a simulated attack-button animation.
- Added physical stroke qualification, initial-acceleration coverage, reversal/backhand rearming and once-per-creature-per-stroke contact. A resting blade, tiny jitter or walking a stationary weapon into an enemy should not repeatedly cause damage.
- Added session, equipment, pose-age, tracking-jump and position checks, plus cancellation during menus, casting, potion handling, recentering and equipment changes. Accepted native contacts can trigger haptic feedback.
- Kept right-trigger manual attacks as a fallback, with guards against simultaneous gesture/button duplicate attacks.
- Corrected render-to-native ground coordinates for elevated dungeon terrain. Cave swings now compare the saved rendered pose and native player/enemy positions consistently without removing the teleport/warp guard.
- Refreshed native local-player identity after area changes and recovered abandoned callback tickets, restoring wrist data and allowing subsequent interaction work to run after transitions.
- Unified the input and posed-contact angular-motion limits so valid fast swings are not discarded before collision recognition. Added recovery checks across area interruptions and several sampling rates.
- Changed free melee to remain available while a wheel skill is selected. Selecting a spell changes the trigger action, while ordinary physical swings still work when the player is not casting or in another exclusive interaction.

- Added an optional contact_hits policy that removes local-player random accuracy and block/dodge/avoid/evade misses against creatures. Native damage, resistances, immunity, resource rules and eligible effects remain; incoming and player-versus-player rules are not overridden.

Important limits: colliders are approximate, main-hand only, and physical stroke timing replaces the basic attack's animation cadence. Independent off-hand damage, physical shield blocking/parrying and attack-speed-stat scaling are not complete. contact_hits intentionally changes combat balance. The original cave-coordinate fix was owner-confirmed; the newest selected-skill/fast-swing follow-up has automated coverage and logged cave contacts, but not exhaustive headset confirmation of every scenario.

6. Two-hand bow controls and ordinary-arrow flight

- Added a nock/grab/draw/release gesture using both tracked hands. The draw line sets aim; minimum draw/time and trigger hysteresis distinguish a deliberate shot from a brief squeeze or tracking loss.
- Added cancellation for menus, equipment changes, tracking loss and quick-cast takeover, preventing a canceled draw from firing.
- Added immediate native ordinary-arrow creation with normal ammunition and damage handling, plus a separate client visual that does not spend a second arrow.
- Added straight 3D basic-arrow flight, swept segment contact against creature capsules and native grid barriers, pierce deduplication, native lifetime and preserved damage/proc behavior. Launch pose is frozen so later hand/head motion cannot steer an arrow in flight.
- Added draw-dependent flight speed and configurable basic-arrow speed. Longer draws affect speed within the implemented range, without a new damage multiplier.
- Added continuous string/limb/nocked-arrow posing for recognized bow rigs, initially the hunter's bow, with attachment/ownership checks and nock/draw/release haptic requests. Unknown rigs retain fallback guides and native animation.

The physical flight path covers ordinary basic arrows. Special, explosive, elemental and guided missiles, crossbow loading, gravity, exact terrain/room behavior and broad bow-rig alignment remain incomplete or require further checks. Native skill controls remain available for unsupported special attacks.

7. Spell wheel, skill assignment and quick casting

- Added a themed eight-slot radial wheel, native localized skill/item icons and names, highlighted selection, a basic WEAPON mode, and an equipment/belt page.
- Added an in-VR skill editor: choose a slot, browse learned or item-granted active skills, assign or clear it, and wait for confirmed native readback. Both native weapon-set binding banks and all four alternate slots are represented.
- Read the native controller binding table directly so mouse/keyboard menu use cannot make the wheel read the wrong binding mode. Assignments continue to use the game's own persistence and duplicate-slot rules.
- Added eight quick-cast chords using Left Trigger plus a face button, with Left Grip for the second group. The wheel displays their VR bindings; repeat/channel behavior and native costs are retained.
- Added ownership guards so quick-cast buttons cannot also open inventory/settings, interact, cancel or grab a potion. Selection itself cannot cast, and held buttons must be released across context changes.

- Improved directional spell aiming from the visible casting shaft or hand ray, and suppressed native target snapping only for the active directed cast. Target-only, corpse, self/area and ground-placement skills keep their appropriate native behavior.
- Removed the redundant Journal entry from the wheel. Journal remains available in the native tabbed game menu.

Full 3D flight/contact for every spell family, immediate physical spell spawning, reliable terrain-aware targeting for every ground skill, and complete charge/cooldown readiness indicators are not implemented.

8. Belt potions, wrist status and weapon configurations

- Added a tracked waist belt with four native potion-column grab positions, live contents/counts and the next available potion from each column. Extra equipped-belt rows supply reserves behind those same positions; potions can appear even without an equipped belt if the native slots contain them.
- Added original 3D potion-model/texture extraction from the installed game for all 15 supported variants, including healing/mana tiers and rejuvenation variants. The assets stay local; the release ships extraction code, not those game models.
- Added nearest-slot reach highlighting, either-hand grip grab, tracked held bottles and release-to-return. Drinking checks the bottle opening near the mouth, tilt and dwell time before one native consumption request.
- Added item-identity/refill/reorder revalidation, confirmation-based consumption feedback and a wheel quick-use alternative. Refused requests are not automatically retried.
- Added back-of-left-wrist health/mana orbs, upright values and low-resource feedback using fresh native snapshots. Corrected loss of those values after dungeon transitions.
- Added seated belt placement and adjustable belt height/forward reach, including closer placement for the owner’s reported reach issue.
- Added dominant-hand trigger-over-either-shoulder weapon-set swapping. The swap waits for native set confirmation and consumes the trigger so it cannot also attack or repeat while held.
- Added a dominant-hand setting for the shoulder gesture, plus wheel Weapon I/II and stow/draw choices as accessible alternatives. Removed native-transport hip weapon anchors and the physical right-stick gameplay swap binding.

Potion models currently use their own depth buffer, without full world/arm occlusion or liquid simulation. Physical drinking, mixed/refilled belts, weapon combinations and hand preferences are not exhaustively verified. Choosing a left shoulder-swap hand does not remap all combat controls.

9. Native game menus and VR settings

- Added laser selection and click-drag interactions for inventory, stash, merchants and other native panels, with stable panel-to-pointer mapping and protection against accidental click-through or nested-menu closure.
- Added native controller menu navigation alongside the laser: face buttons, both sticks and stick clicks, grips as bumpers, triggers and supported Menu/Start input follow the game’s displayed prompts.
- Added deliberate input handoff. A stick/face-button action selects native navigation; idle ray jitter cannot steal selection. Releasing native controls and deliberately re-aiming returns to laser interaction.

- Retained Right Trigger laser selection/dragging; moved laser context-click to Left Grip + Right Trigger; retained Left Trigger Ctrl-click and Right Grip Shift modifiers. A modifier change cannot switch a drag's mouse button midway through the hold.
- Made the left primary button open the tabbed menu during gameplay and act as native X inside menus. Added explicit inventory X/drop and full held-right-stick sort support. Gameplay weapon swapping remains on the shoulder gesture or wheel.
- Added per-control release gates and menu-generation checks so held or stale gameplay/menu reports cannot leak into a new context. Native UI bypasses gameplay turn/attack translation; mod settings/calibration retain exclusive VR input.
- Added UI-only journal capture, spatial placement/pinning support and a normal-panel fallback when capture is unavailable. Access remains through the native tabbed menu; extra decorative journal pages are not complete.
- Added an in-headset settings panel with Controls, VR and Video tabs. It exposes motion combat, seated belt placement, shoulder-swap hand, turn mode/speed, belt reach, foveation, inherited sky/fog/ceiling effects and terrain/object draw distances.
- Added Apply/cancel behavior, live reload, original-INI backup, conflict detection, save feedback, height reset and body recalibration. Settings remain accessible when motion combat is off and while XR is active at the frontend.
- Fixed the Akara dialogue freeze by queueing window activation outside the locked render/input callback, with expiry/cancellation for stale focus requests. Subsequent dialogue sessions opened/closed and returned to gameplay.

Basic native menus, laser dragging and matching stash backgrounds were owner-confirmed. The newest inventory X/drop and held-stick sort revision still awaits a fresh headset check. Native game Video settings remain responsible for DLSS/textures/shadows; full control rebinding and an in-mod resolution override are not included.

10. Installation, launchers and Steam library integration

- Added a reversible installer that checks the supported Steam game build and verifies pinned official D2RLoader/ReShade downloads. It backs up replaced files and saves, and does not replace the original D2R.exe or install an automatically loaded DXGI proxy.
- Added local native icon and potion preparation, with bounded Steam archive/texture/model reading and fallback labels/previews when optional caches are unavailable. Licensed reader/decoder components retain their attribution.
- Added Windows launch wrappers and separate SteamVR/VDXR launchers. The wrapper consumes the mod's VR flags, routes explicit VR launches to D2RLoader, and keeps normal Steam Play on the original stock executable.
- Added per-launch OpenXR manifest selection, including an explicit VDXR override. This switches this game's runtime without changing the system's global OpenXR registration; missing required runtimes stop the VR launch rather than silently switching routes.
- Added Install-Windows.cmd and a guided Windows setup wizard. It selects Steam Frame/Quest 3/Quest 2/other SteamVR, offers SteamVR/Virtual Desktop/both for Quest, finds separate Steam libraries, selects a Steam account and confirms the saved-games location.

- Added a private Python bootstrap from python.org with a pinned archive checksum and publisher-signature validation. Players do not need a separate Python installation or a system PATH change.
- Added automatic D2R launch options and matching non-Steam library shortcuts, including the SteamVR VR-library flag and a VDXR shortcut that does not request SteamVR. Repeated setup updates existing entries without duplicates.
- Added Steam-closed checks, exact binary shortcut round trips, narrow launch-option edits, original configuration backups, external-change detection and rollback after a partial configuration-write failure. Setup never forcibly closes Steam or the game.
- Added update/repair mode that retains calibration, video and existing ReShade configuration, plus a shortcuts-only mode. Restore preserves current character progress; settings/backups remain available for removal.

11. Diagnostics, reproducibility and player documentation

- Added bounded stereo-cycle, CPU-hook, memory, XR wait/locate/input/acquire/copy/release/submit and optional GPU timing captures. Sampling stops automatically; routine play avoids screenshot readback and raw pose logging.
- Added paired-eye capture requests and remote capture controls to inspect body, label and rendering problems while the player keeps a natural posture.
- Added performance-report and SteamVR-stream-history tools, including overlapping-history deduplication, timestamp-wrap handling, missing-data preservation and separate delivery/encode/decode stages.
- Added native runtime probing, signature/ABI checks and regression fixtures for math, input, combat, UI, GPU formats, settings, installation and Steam configuration. Current recorded totals are 34 native fixtures and 34 Python tests.
- Added deterministic Windows build/toolchain setup and source/hash manifests for release packaging. The explicit package allowlist excludes game assets, extracted caches, saves, logs, dumps and local backups.
- Added player-facing Windows instructions, full Quest/Steam Frame controller mappings in the README and ZIP, ownership/no-game-assets notices, separate runtime launch directions, shortcut instructions and upgrade/removal guidance.
- Added private Windows preview releases with checksum assets. The v0.3.3 package includes the setup wizard and latest inventory controls; the installed local DLL matches its packaged build.

12. Validation and present boundaries

Area	Recorded status
Steam Frame / Windows / SteamVR	Offline gameplay confirmed; torso clipping, original cave-coordinate fix, calibration and shoulder behavior were covered by owner feedback.
Quest 3 / SteamVR and VDXR	Both routes have gameplay confirmation on preceding builds; VDXR color correction explicitly confirmed. This is not every feature on every later build.

Area	Recorded status
Menus / dragging / stash stereo	Owner-confirmed on the preceding menu build. Latest inventory hold/drop update is installed and passes fixtures; fresh headset confirmation is pending.
Setup wizard	Routing, upgrades, idempotence, backups, rollback and races checked with isolated fixtures. Clean extracted-package bootstrap/integrity check passed. Full fresh-PC install trial remains pending.
Performance	Windows feels very good to the owner. No completed controlled Windows baseline/current benchmark or proven general cross-OS speedup.
Quest 2 and unsupported game builds	Quest 2 hardware check pending. Version-sensitive native signatures stay guarded; this release targets Steam 3.3.93787.

The preview is for local offline single-player. A stock launch route is provided for ordinary desktop/multiplayer use after fully closing VR; this document does not claim that running the injected mod online is supported. PR #3 is merged into main at dcbfcf57ada9eaff7d1d7f81a5305301a0a89f81. Its source tree matches the released feature snapshot, and the local main checkout and installed DLL are up to date. The repository and release remain private.

Other limits include approximate body/weapon/creature fitting, incomplete physical off-hand/shield/spell-family features, limited class/item/boss coverage, and unresolved broad DLSS clarity. Historical Linux streaming resets and large simulation stalls were investigated separately; they are not presented as universally cured by the Windows build. Signature-guarded unavailable native paths remain disabled rather than forced on.

Controller reference

The complete current mappings below are taken from CONTROLS.md at the documented source snapshot. They cover both new interactions and retained controls needed to use them.

Windows preview for SteamVR and Virtual Desktop (VDXR). These controls use two tracked VR controllers; an Xbox controller is not required.

Button names

Hand / button	Quest 2 / 3 label	Steam Frame label
Right primary / secondary	A / B	A / B
Left primary / secondary	X / Y	A / B
Trigger	Index trigger	Index trigger
Grip	Grip button	Grip input

Hand / button	Quest 2 / 3 label	Steam Frame label
Stick click	Press the thumbstick	Press the thumbstick

Left primary below means Quest Left X / Frame Left A. Left secondary means Quest Left Y / Frame Left B.

Gameplay

Action	VR control
Walk	Left stick
Snap turn	Right stick left / right
Interact / pick up / talk	Right A while aiming at the target
Open the tabbed game menu	Left primary; select Inventory, Character, Skills or Journal in its tabs
Spell wheel	Hold left secondary and turn your left palm toward you; select with right laser + trigger
Assign wheel skills	Wheel → EDIT SKILLS → slot → learned skill
Select basic weapon mode	Choose WEAPON in the wheel center, then release the trigger
Free melee swing / thrust	Move your equipped melee weapon through an enemy; also works with a wheel skill selected
Manual attack / selected skill	Right trigger; hold for the skill's native repeat behavior
Bow	Bring the right hand to the bowstring, hold right trigger, draw back, then release
Cancel a grab / bow draw	Right B, then release the inputs
Swap weapon sets I / II	Reach the swap hand over either shoulder and press its trigger; right hand by default, adjustable in VR settings
Seated weapon selection / stow	Wheel → EQUIPMENT / BELT → Weapon I, Weapon II or Stow/Draw
Grab a belt potion	Reach to a bottle and hold that hand's grip
Drink	Keep grip held, bring the bottle opening to your mouth and tip it; release an unused bottle to return it
Quick potion	Wheel → EQUIPMENT / BELT → select a potion with right trigger
Wrist health / mana	Look at the back of your left wrist, like checking a watch

Action	VR control
Map	Left stick click
Reset height	Left Grip + left stick click, during gameplay
VR settings	Left Grip + left secondary; also VR SETTINGS in the wheel
Calibrate height / arm span	VR settings → Calibrate body; stand straight in a T-pose, release both triggers, then hold both together until Saved
Recenter	PC F11
Save stereo images	PC F8

Free swings pause during casting, menus and potion handling. Selecting a wheel skill changes the right-trigger action; its quick-cast binding stays available. The belt shows potions in the game's belt slots even without an equipped belt.

Quick-cast shortcuts

Hold Left Trigger, aim the right hand, then press the assigned face button. Add Left Grip for slots 5–8. Assign the eight slots through EDIT SKILLS; the wheel shows each slot's VR shortcut. Release the face button before changing slots or the grip modifier.

Wheel slot	Quest shortcut	Steam Frame shortcut
1	Left Trigger + Right A	Left Trigger + Right A
2	Left Trigger + Right B	Left Trigger + Right B
3	Left Trigger + Left X	Left Trigger + Left A
4	Left Trigger + Left Y	Left Trigger + Left B
5	Left Trigger + Left Grip + Right A	Left Trigger + Left Grip + Right A
6	Left Trigger + Left Grip + Right B	Left Trigger + Left Grip + Right B
7	Left Trigger + Left Grip + Left X	Left Trigger + Left Grip + Left A
8	Left Trigger + Left Grip + Left Y	Left Trigger + Left Grip + Left B

Menus: laser and native buttons

Both methods work in the main menu and native game panels, including inventory, stash and merchants. Keep the D2R window focused on the PC.

Use a stick or face button to start native button navigation and follow the game's displayed prompts. The idle laser will not move your native selection. To return to the laser, release the native controls and deliberately move your aim across the panel. A small hand tremor will not change methods.

Laser controls

Action	VR control
Hover / point	Aim the right controller's laser
Select / drag an item	Right trigger; hold while moving the item and release to place it
Context / alternate click	Left Grip + Right Trigger (hold grip first)
Native Ctrl-click shortcut	Hold Left Trigger, then click with Right Trigger; the game chooses drop, sell or transfer for the current panel
Native Shift modifier	Hold Right Grip while using the laser
Back / close	Right B, or point and click the panel's close/back button
Tabs / page controls	Point and click the native on-screen controls

Dragging keeps the item held if the ray briefly leaves the panel. Releasing the trigger ends the drag at the last valid pointer position. Tracking or desktop focus loss releases the injected mouse button; release the trigger before starting again. Pressing or releasing Left Grip during a drag will not convert it into a context-click.

Native menu prompts

Game's displayed input	Quest control	Steam Frame control
A	Right A	Right A
B / back	Right B	Right B
X	Left X	Left A
Y	Left Y	Left B
Left stick / right stick	Left stick / right stick	Left stick / right stick
LS / RS click	Left stick click / right stick click	Left stick click / right stick click
Inventory: X / drop selected item	Left X; hold if the native prompt requires it	Left A; hold if the native prompt requires it
Inventory: hold RS / sort	Press and hold right stick click	Press and hold right stick click
LB / RB	Left Grip / Right Grip	Left Grip / Right Grip
LT / RT	Left Trigger / Right Trigger	Left Trigger / Right Trigger
Menu / Start, where available to the game	Controller menu button	Controller menu button

Native menu actions use the same buttons in the main menu, tabbed menu, stash, inventory and merchant panels. Left primary opens the tabbed menu during gameplay; inside a native menu it performs the displayed X action. Left Grip + left secondary stays reserved for VR settings. Right-stick click does not swap weapons during gameplay. Use the shoulder gesture or EQUIPMENT / BELT wheel for weapon configurations.

Quest controllers do not need a separate D-pad or Xbox controller: use sticks for native navigation and the laser for the on-screen controls. When using the laser, Right Trigger is a mouse click and the grips/triggers above provide its modifiers; once native navigation is selected, they perform LB/RB/LT/RT. Release buttons, triggers and sticks when opening or closing menus so a held input cannot also act in the next screen or in gameplay.

The mod's own settings and calibration panels use their VR laser controls: right laser + trigger to select, Right B to close/cancel. Native gamepad input does not pass through those panels.

Source references

Prepared from the repository diff against the pinned upstream commit, current source, retained upstream credits and recorded validation evidence. Historical handoff/task notes contain intermediate states; the feature descriptions above follow the current source and later confirmations.

Upstream base: BodyWalkVR D2R VR

Credits and third-party notices

Retained upstream documentation

Native OpenXR and rendering notes

Motion gameplay implementation and limits

Body calibration and shoulder swaps

Controller mappings

Performance evidence and limits

Windows state and fixes

Quest transport evidence

Latest setup/inventory evidence

Windows install guide

Private v0.3.3 Windows preview release